

AI Investment and National AI Capability

What AI Investment Does and Does Not Buy: Evidence From Ten National AI Ecosystems

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Abstract

Governments and investors increasingly treat artificial intelligence investment as a proxy for national AI strength. This paper examines whether that assumption holds by comparing OECD AI venture-capital investment with Stanford HAI's Global AI Vibrancy indicators across the ten countries leading Stanford's 2024 ranking. Because Stanford's composite score incorporates private investment directly through its Economy pillar, this paper removes that pillar and constructs an independent Non-Economic AI Ecosystem Score spanning six dimensions: R&D, responsible AI, talent, policy and governance, public opinion, and infrastructure. Both datasets are benchmarked to 2024.

Across the ten-country sample, AI investment has a positive but statistically insignificant association with overall ecosystem strength (Spearman $\rho = 0.41$, $p = 0.24$). The relationship is stronger when capability is measured by ecosystem breadth: higher-investment countries tend to have capability distributed more evenly across the six dimensions ($\rho = -0.64$, $p = 0.048$). The results suggest that investment may be more informative about the breadth of an AI ecosystem than about its aggregate level of capability. Countries with substantially different investment levels can reach similar ecosystem scores through different combinations of research, talent, governance, infrastructure, and other capabilities.

The analysis also separates ecosystem capability from strategic autonomy. External evidence shows that countries can develop substantial domestic AI research, talent, and infrastructure while remaining dependent on foreign firms for critical technologies, particularly advanced semiconductors and cloud infrastructure. AI investment is therefore an incomplete measure of national capability and cannot, by itself, establish technological sovereignty.

1. Introduction: From AI Investment to AI Capability

Artificial intelligence has increasingly become a domain of national competition. The shift was particularly visible in the aftermath of DeepMind's AlphaGo victory over Lee Sedol in March 2016, which demonstrated the rapid progress of machine learning in a field with considerable symbolic importance in China. The following year, China's State Council published the *New Generation Artificial Intelligence Development Plan*, establishing a national objective of becoming a global AI leader by 2030. The plan did not create the international AI race by itself, but it reflected a broader transition in which AI development became increasingly associated with national technological and economic power.

That competition intensified after OpenAI released ChatGPT in November 2022. Venture investment in AI companies accelerated, foundation-model development attracted unprecedented private capital, and large technology companies began redirecting substantial capital expenditure toward model training, inference, and data-center infrastructure. Governments likewise began treating compute, semiconductor supply, AI talent, research capacity, and domestic model development as strategic assets. In this environment, the amount of money flowing into AI has become one of the most visible indicators of national ambition. Countries attracting large amounts of private investment are often assumed to be building correspondingly large amounts of national AI capability.

The assumption is plausible. Capital can finance research, infrastructure, talent acquisition, commercialization, and the development of firms capable of deploying AI at scale. But investment is not capability itself, and countries differ in how capital interacts with the institutions, talent pools, research systems, and infrastructure they already possess. A country can invest heavily in a narrow part of the AI ecosystem while remaining weak elsewhere; another can attract considerably less private capital while developing strong research institutions, talent pipelines, or regulatory capacity. The relevant question is therefore not simply whether countries that invest more become stronger, but what aspects of national capability investment is actually associated with.

This paper examines two related questions. First, is higher AI investment associated with stronger national AI ecosystems? Second, is investment associated with the breadth of those ecosystems—whether capability is distributed across multiple dimensions rather than concentrated in one or two? To examine these questions, the paper combines 2024 OECD AI venture-capital data with Stanford HAI's 2024 Global AI Vibrancy indicators for the ten countries at the top of Stanford's ranking. Because Stanford's overall score already incorporates investment through its Economy pillar, that pillar is excluded from the capability measure. The analysis therefore compares investment against six other dimensions of ecosystem development rather than against a measure that partially contains investment by construction.

The results point to a more differentiated relationship than the investment narrative suggests. Investment has a positive but statistically insignificant relationship with aggregate ecosystem strength within the sample, while its relationship with ecosystem breadth is stronger and statistically significant. The finding does not establish that investment causes broader capability; the cross-sectional design cannot distinguish that mechanism from reverse causality or common underlying factors. It does, however, suggest that investment and capability capture different aspects of national AI development. The final section extends this distinction to strategic autonomy, showing why even a broad and well-developed ecosystem should not automatically be interpreted as technologically sovereign.

2. What Do We Mean by AI Capability?

The phrase “AI capability” is used broadly in policy discussions, often encompassing financial investment, research output, talent, infrastructure, institutional capacity, and technological independence within a single concept. These dimensions are related but not interchangeable. A country can have a large AI investment market without possessing equally broad research or infrastructure capacity, while a country can have strong research institutions or talent without attracting comparable levels of private

capital. Separating these concepts is therefore necessary before asking what investment actually tells us about national AI strength.

AI capital refers to the financial resources flowing into AI-related companies and activities. This paper measures AI capital using country-level venture-capital investment reported by the OECD, based on Preqin deal data. Venture investment is an important component of the AI ecosystem because it provides financing for firms, technologies, and commercialization, but it represents a flow of financial resources rather than a direct measure of the capabilities those resources produce.

AI ecosystem strength refers to the broader set of conditions supporting AI development. Here it is measured using six dimensions from Stanford HAI's Global AI Vibrancy indicators: R&D, responsible AI, talent, policy and governance, public opinion, and infrastructure. The resulting score captures a wider range of national capabilities than investment alone, although it should not be interpreted as a complete measure of technological capability in every sense.

Strategic autonomy is different again. It concerns the degree to which a country can develop and operate AI without depending on foreign providers for critical technologies. A country can possess strong domestic research, infrastructure, and talent while relying on foreign companies for advanced chips, cloud computing, networking equipment, or foundation models. Strategic autonomy is therefore treated in this paper as a separate dimension rather than as an implied consequence of ecosystem strength.

The distinction among these concepts provides the framework for the empirical analysis. The central question is not whether capital matters to AI development—it clearly does—but whether investment totals provide a useful proxy for the broader capabilities that governments and investors increasingly seek to measure.

3. Data and Methodology

This paper uses a descriptive cross-sectional design to examine the relationship between AI investment and national ecosystem characteristics. It does not attempt to establish causality. In particular, a positive association between investment and capability could reflect capital contributing to capability, stronger ecosystems attracting investment, or other factors that influence both. The purpose of the analysis is instead to determine which aspects of ecosystem development are most closely associated with differences in investment across the countries examined.

3.1 Data sources

AI venture-capital investment by country and year comes from the OECD AI Policy Observatory, which aggregates Preqin deal data (OECD.AI, 2026). National AI ecosystem indicators come from Stanford HAI's Global AI Vibrancy Tool, which scores countries across seven pillars using min-max normalized indicators that are aggregated into pillar scores and an overall composite (Stanford HAI, 2026a). The two datasets are aligned to 2024 so that the investment and ecosystem measures describe the same period.

The OECD data capture venture investment rather than the full universe of public and private AI expenditure. This distinction is important when interpreting countries such as China, where state funding and other forms of public investment may play a larger role than venture capital. Stanford's indicators likewise capture multiple dimensions of an ecosystem but do not constitute a direct measure of technological independence or control over critical supply chains.

3.2 Removing circularity

Stanford's Economy pillar incorporates Total AI Private Investment and Newly Funded AI Companies, each weighted at 50 percent within that pillar. Using Stanford's overall composite as the measure of

ecosystem strength would therefore introduce structural overlap with the investment variable being tested. A country with high private investment would receive a higher overall score partly because of that same investment.

To avoid this circularity, the Economy pillar is removed entirely. The paper constructs a Non-Economic AI Ecosystem Score as the equal-weighted average of the remaining six pillars:

$$\text{Ecosystem Score} = \frac{R\&D + Responsible\ AI + Talent + Policy\ \&\ Governance + Public\ Opinion + Infrastructure}{6}$$

The resulting score is an analytical measure constructed for this paper rather than a Stanford-defined index. It should therefore be understood as a measure of the breadth of ecosystem dimensions represented in Stanford's data, rather than as a universal measure of technological capability.

3.3 Measuring breadth

Aggregate strength does not capture how capability is distributed within an ecosystem. To measure this distribution, the paper uses a Herfindahl-Hirschman Index (HHI), a concentration measure commonly used in industrial organization to assess market concentration. Here the HHI is applied descriptively to the six capability dimensions within each country rather than to market shares across firms or countries.

For each country, the six pillar scores are converted into shares of the country's total Non-Economic Ecosystem Score. These shares are squared and summed:

$$HHI = \sum_{i=1}^6 \left(\frac{\text{pillar score}_i}{T} \right)^2$$

A lower HHI indicates that capability is distributed more evenly across the six dimensions, while a higher HHI indicates greater concentration in a smaller number of dimensions. With six equally weighted dimensions, the theoretical minimum is 1/6, or approximately 0.167. The measure therefore complements the Ecosystem Score: the first captures the level of capability, while the second captures its internal distribution.

3.4 Correlation measure

The relationship between investment and the two ecosystem measures is evaluated using Spearman's rank correlation coefficient. Spearman correlation is appropriate for the small sample and avoids assuming a linear relationship between investment and capability. It measures whether countries tend to occupy similar or opposing positions in the rankings of the two variables. The reported p-values indicate the statistical significance of the observed rank association within the ten-country sample.

3.5 Country sample

The sample consists of the ten countries ranked highest by Stanford HAI's 2024 Global AI Vibrancy Index: the United States, China, India, South Korea, the United Kingdom, Singapore, Spain, the United Arab Emirates, Japan, and Canada. The sample is therefore intentionally focused on countries already identified as leading AI ecosystems rather than on a representative cross-section of the global economy.

This selection introduces an important limitation because Stanford's overall ranking includes the Economy pillar. The countries in the sample were therefore selected using a measure that partially incorporates investment-related indicators. The analysis does not attempt to treat the sample as independent of investment; instead, it asks whether investment continues to explain meaningful variation

in ecosystem characteristics after the investment-linked Economy pillar has been removed from the capability measure.

Stanford's own tool provides a useful sensitivity check. When the Economy pillar is excluded from the ranking, nine of the ten countries retain broadly similar positions, while the United Kingdom moves from fifth to ninth. The result indicates that investment-linked indicators can materially affect individual country rankings even within a group of already-leading AI economies.

3.6 Limitations

The analysis has three principal limitations. First, the sample contains only ten countries and is selected from Stanford's highest-ranked economies, limiting statistical power and generalizability. Second, China's and Japan's Talent scores use Stanford's own cross-country median imputation of 34.2 where underlying indicators were unavailable. Third, the analysis is cross-sectional. It cannot determine whether investment produces broader capability, whether stronger ecosystems attract more investment, or whether both are driven by other factors. These limitations mean that the correlations should be interpreted as descriptive evidence of association rather than as estimates of a causal effect.

4. Where Is AI Capital Going?

The first feature of the investment data is its extreme concentration. Figure 1 shows the development of AI venture investment across countries from 2012 through the most recent available year, while Figure 2 isolates the 2024 cross-country distribution used in the remainder of the analysis.

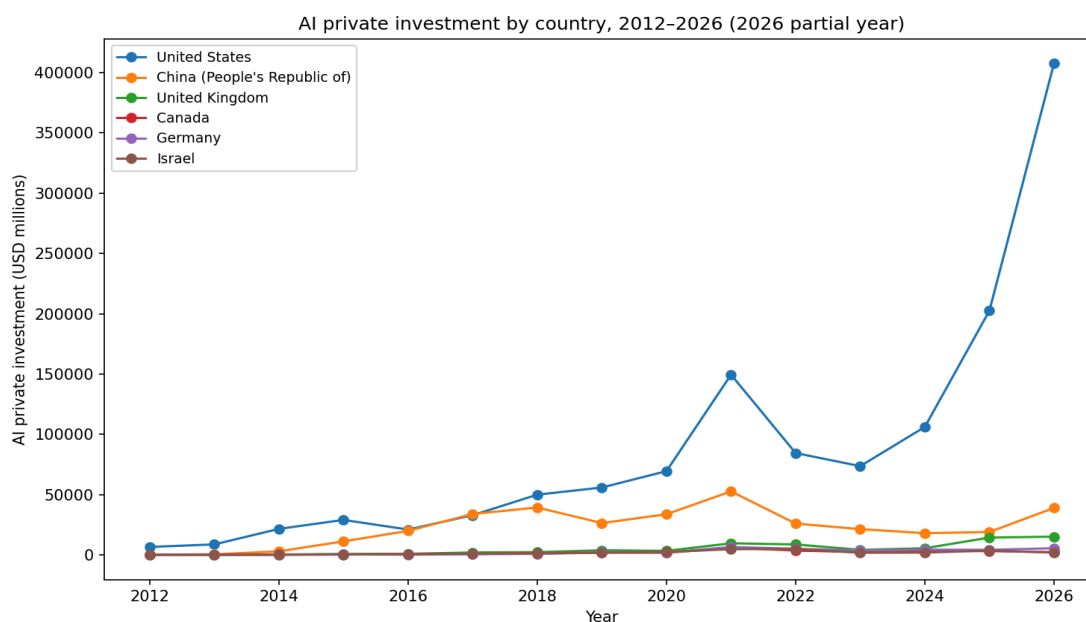


Figure 1. AI venture investment by country, 2012–2026 (OECD/Prequin data). 2026 is a partial year.

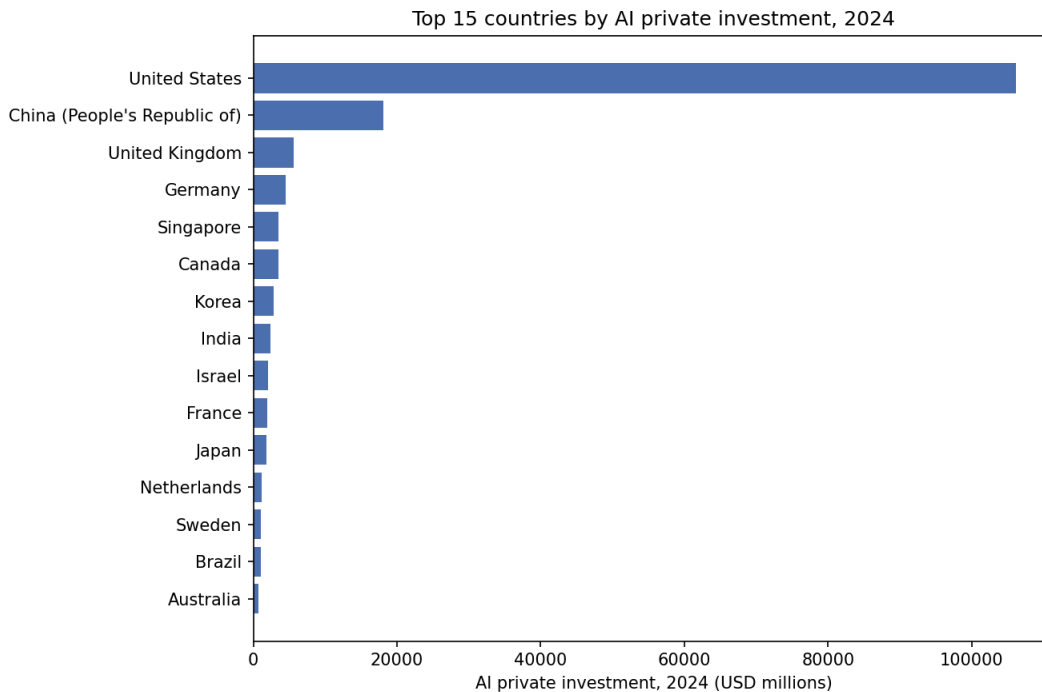


Figure 2. Top 15 countries by AI venture investment, 2024.

Investment accelerates sharply after 2022, coinciding with the rapid commercialization of generative AI and the expansion of investment around foundation models, AI infrastructure, and related applications. By 2024, however, the global distribution remains highly uneven: the United States accounts for \$106.2 billion in AI venture investment, compared with \$18.1 billion for China and \$5.6 billion for the United Kingdom.

The scale of this difference is important for interpreting the ecosystem results. The United States attracted almost six times as much AI venture investment as China and more than thirty times as much as India, yet the distribution of ecosystem scores is substantially narrower than the distribution of investment. Several countries receiving a fraction of U.S. investment nevertheless record ecosystem scores within a relatively small range of one another. Investment is therefore highly concentrated, while the broader measures of ecosystem development are not concentrated to the same degree. This divergence provides the empirical basis for asking whether investment is primarily associated with the overall level of capability or with some other structural characteristic of an ecosystem.

5. Investment and Overall Ecosystem Strength

The first empirical test asks whether the countries receiving the most AI investment also exhibit the strongest ecosystems across the six non-economic dimensions. The relationship is positive but not statistically significant: the Spearman correlation between 2024 AI investment and the Non-Economic AI Ecosystem Score is $\rho = 0.41$ ($p = 0.24$). The direction of the relationship is therefore consistent with the idea that capital and ecosystem strength tend to move together, but the evidence is not strong enough within this sample to treat investment as a reliable predictor of aggregate ecosystem strength.

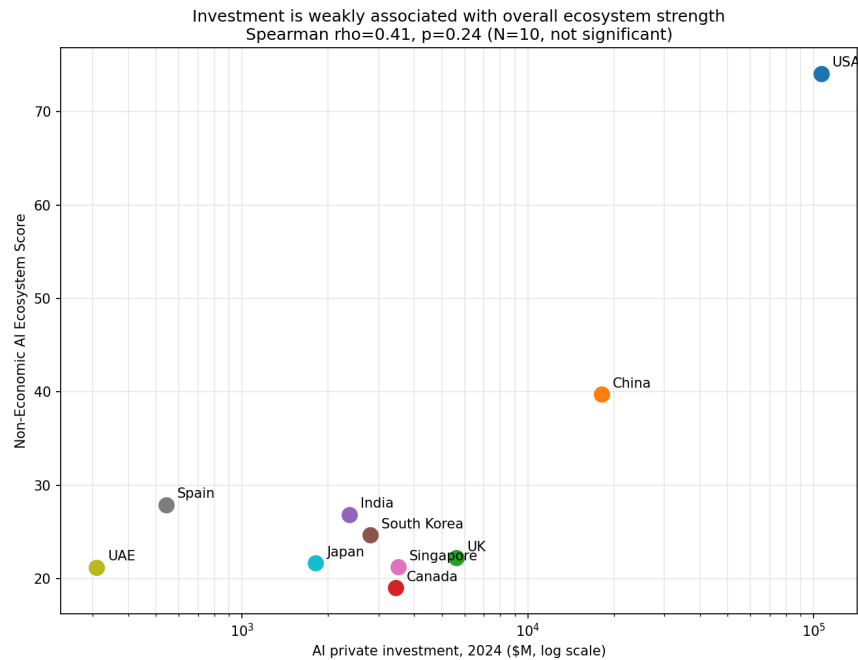


Figure 3. AI investment vs. Non-Economic AI Ecosystem Score, ten-country sample, 2024.

The country-level distribution helps explain the result. The United States ranks first on both investment and ecosystem strength, while China ranks second on both, establishing much of the positive direction in the correlation. Below these two countries, however, investment levels diverge substantially without corresponding differences in ecosystem scores. Spain receives only \$542 million in AI venture investment but records an ecosystem score of 27.9, higher than South Korea, Singapore, the UAE, Japan, and Canada. Singapore receives \$3.5 billion—more than six times Spain's investment—while recording an ecosystem score of 21.3. India receives \$2.4 billion but scores 26.9, while Canada receives \$3.4 billion and scores 19.0. The data therefore show no simple conversion rate between capital and aggregate ecosystem strength.

This pattern does not mean that capital is unimportant. Rather, it indicates that investment represents only one input into national ecosystem development. The capacity to translate financial resources into research, talent, infrastructure, governance, and other forms of capability depends on the institutions and technological base surrounding that investment. Conversely, some capabilities can be developed through channels that are not captured by venture investment, including public research funding, education systems, immigration policy, and government-led institutional development. The cross-country comparison therefore provides little support for treating venture investment alone as a sufficient proxy for national AI strength.

6. Investment and Ecosystem Breadth

The relationship becomes stronger when the question shifts from how much capability a country has to how evenly that capability is distributed. The Spearman correlation between AI investment and HHI is -0.64 ($p = 0.048$). Because lower HHI values indicate a more even distribution across the six dimensions, the negative relationship means that countries with higher investment tend to have broader ecosystems. This association reaches conventional statistical significance despite the small sample, although the result should still be interpreted cautiously given the limited number of observations.

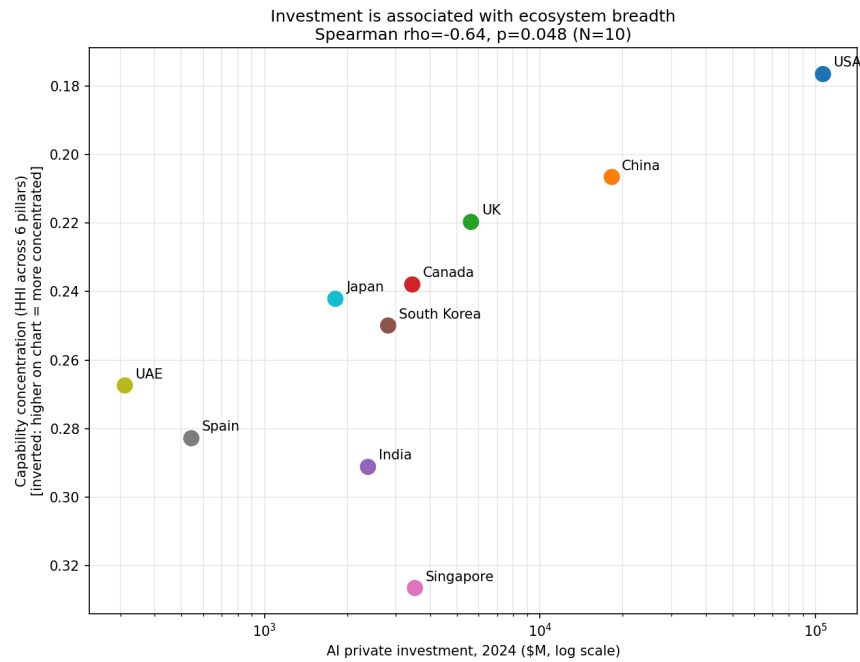


Figure 4. AI investment vs. capability concentration (HHI), 2024. Higher on the chart (inverted axis) indicates a more concentrated, less broadly developed ecosystem.

Table 1 reports the full country-level data underlying both results, including each country's pillar scores, Ecosystem Score, HHI, and 2024 investment total. All figures in the table are for the same year.

Country	R&D '24	Resp. AI '24	Econ. '24	Talent '24	Policy '24	Pub. Opin. '24	Infra '24	Ecosys. Score	HHI	Inv.2024 (\$M)
United States	66.8	100.0	100.0	54.0	67.9	59.0	96.9	74.1	0.176	106,165
China	67.2	39.9	8.6	34.2*	3.5	52.5	41.2	39.7	0.206	18,140
United Kingdom	9.4	9.8	7.2	33.3	43.8	19.9	17.1	22.2	0.220	5,609
India	33.1	6.1	3.8	60.4	5.2	52.4	4.1	26.9	0.291	2,373
South Korea	6.8	2.1	2.8	29.3	26.4	55.6	27.8	24.7	0.250	2,806
Singapore	6.3	5.6	2.0	62.0	26.9	26.0	0.8	21.3	0.326	3,512
Spain	4.4	0.1	0.8	32.3	71.3	32.0	27.2	27.9	0.283	542
UAE	1.8	1.1	1.0	43.5	14.1	35.0	31.7	21.2	0.267	310
Japan	6.5	5.2	2.2	34.2*	10.0	39.0	35.0	21.7	0.242	1,807
Canada	7.9	8.0	3.5	44.7	16.4	16.1	21.1	19.0	0.238	3,440

Table 1. Country-level data, all figures for 2024. Pillar scores are min-max normalized (0–100) per Stanford HAI's Global AI Vibrancy Tool. *China and Japan Talent values are Stanford's own median-imputed figures (see Section 3). Ecosystem Score and HHI as defined in Section 3. Investment figures are OECD/Prequin totals, USD millions.

The country-level data illustrate the difference between strength and breadth. The United States combines the highest investment level with strong performance across every non-economic pillar and consequently has the lowest HHI in the sample, at 0.176. China is also relatively broad, with an HHI of 0.206, despite receiving only a fraction of U.S. investment. At lower investment levels, the ecosystem structures become more concentrated. Singapore's strength is particularly pronounced in Talent, Spain's in Policy and Governance, and India's in Talent and R&D. The UAE similarly derives a large share of its ecosystem

score from Talent, Public Opinion, Infrastructure, and Policy. These countries are not necessarily weak ecosystems; rather, their capability is distributed differently.

The breadth measure therefore captures something that the aggregate score obscures. Spain's ecosystem score of 27.9 is higher than Singapore's 21.3, but the composition of those scores is fundamentally different. Spain's position is strongly supported by Policy and Governance, while Singapore's is strongly supported by Talent. India reaches 26.9 through a combination of R&D and Talent despite much lower infrastructure and governance scores. Treating these countries as equivalent because their aggregate scores are relatively close would conceal meaningful differences in what each ecosystem has developed.

The statistical association does not establish why this pattern exists. One possibility is that larger pools of investment allow countries to develop several parts of the ecosystem simultaneously. Another is that countries with stronger institutions, research systems, and technology sectors are both better positioned to attract investment and more capable of sustaining broad ecosystems. The data cannot distinguish between these explanations. What the result does show is that investment is more closely aligned with the distribution of capability across dimensions than with the aggregate level of capability itself.

7. Different Models of AI Ecosystem Development

The breadth result suggests that countries do not simply occupy different points on a single scale of AI capability. They also develop different combinations of capabilities. Figure 5 makes this visible by showing the composition of each country's non-economic ecosystem score. The United States has relatively balanced performance across the six dimensions, while other countries derive a much larger share of their scores from particular areas. Examining these configurations helps explain why countries with similar aggregate scores can represent very different forms of AI capacity.

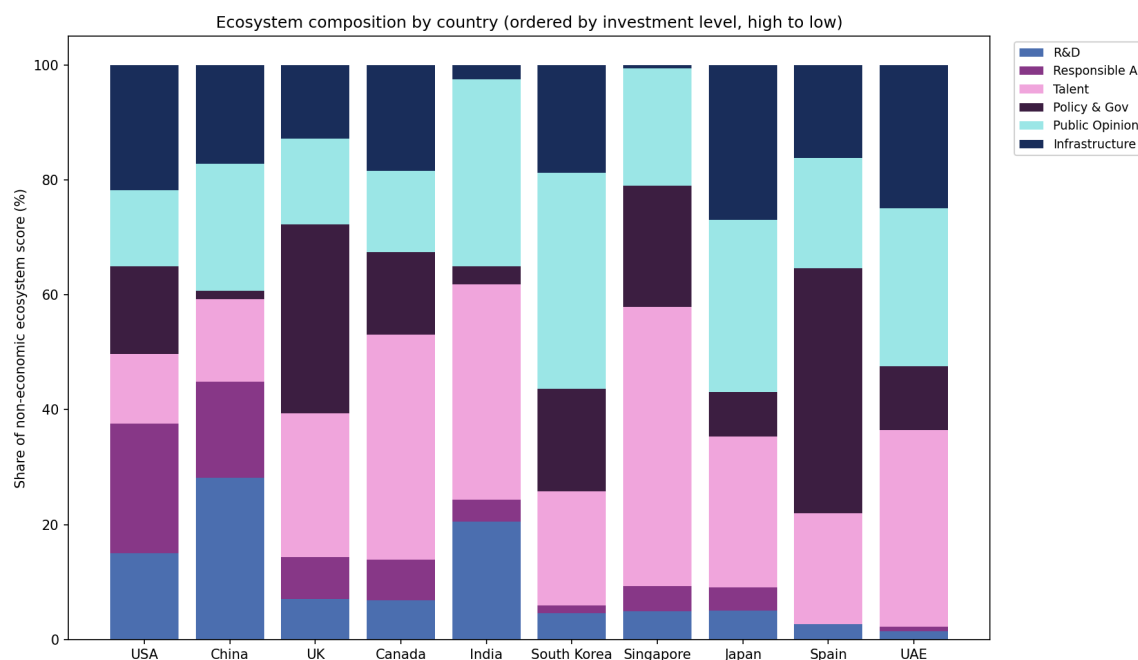


Figure 5. Composition of national AI ecosystems by capability dimension, 2024. Countries are ordered by 2024 AI investment.

7.1 United States: Broad-Based Development

The United States is the clearest example of a broad-based ecosystem. It is above the sample median on every one of the six non-economic dimensions and records the lowest HHI in the sample, at 0.176. Its ecosystem therefore does not depend on a single source of strength: the country combines leading R&D and infrastructure with high scores in responsible AI, talent, policy and governance, and public opinion. This breadth coincides with an exceptionally large investment market, suggesting that the United States has both the financial resources and the institutional and technological base to develop multiple parts of the ecosystem simultaneously.

The data cannot determine whether the scale of U.S. investment caused this breadth. The direction of causality could run partly in the opposite direction, with an already deep technology ecosystem attracting capital. Nevertheless, the United States represents the broadest configuration in the sample and the one most closely aligned with the negative investment-HHI relationship identified above.

7.2 China: Research-Led Development

China illustrates a different model. Its R&D score of 67.2 is second only to the United States, while its Economy score is only 8.6. China's 2024 AI venture investment of \$18.1 billion is roughly one-sixth of U.S. investment, yet its R&D score is almost identical to the U.S. score of 66.8. The comparison demonstrates that private venture investment and research capability can diverge substantially at the national level.

China's research strength may reflect mechanisms that are not captured by venture-capital data, including public research funding, university systems, state-backed programs, and the scale of its scientific workforce. The present dataset cannot determine the relative contribution of these factors, but the divergence itself is important: high national R&D capability does not require a proportional level of private AI investment. China therefore provides a particularly clear example of why venture capital should not be treated as a complete measure of national AI capability.

7.3 Spain: Governance-Led Development

Spain represents another configuration, with Policy and Governance scoring 71.3—the highest value in the sample—while R&D and Responsible AI score only 4.4 and 0.1 respectively. Its overall ecosystem score of 27.9 is therefore driven disproportionately by institutional and regulatory capacity rather than by research or private capital. Spain's AI governance development is consistent with a broader national effort to establish regulatory institutions alongside the implementation of the European Union's AI regulatory framework.

Spain established the Agencia Española de Supervisión de la Inteligencia Artificial (AESIA) in 2023, with the agency becoming operational in June 2024, and developed an AI regulatory sandbox under Royal Decree 817/2023. Its national AI legislation has also continued to develop alongside the EU AI Act (White & Case, 2026; IAPP, 2026). These developments provide external context for the high Policy and Governance score, but they should not be interpreted as evidence that regulation itself produces the score. Rather, Spain demonstrates how a country can develop a relatively strong component of an AI ecosystem without simultaneously developing comparable capability across the other dimensions.

7.4 India, the UAE, Canada, Singapore: Talent-Led Development

India, the UAE, Canada, and Singapore share a different structural characteristic: Talent accounts for a large share of their non-economic ecosystem scores, ranging from roughly 40 percent to more than 60 percent. India combines a Talent score of 60.4 with R&D of 33.1, giving it the strongest research-and-talent combination outside the United States and China in this sample. Its IndiaAI Mission, launched in 2024, includes a FutureSkills component aimed at expanding AI education and research training, including support for doctoral, postgraduate, and undergraduate AI capacity (IMPRI, 2026; Digital in Asia, 2026).

The UAE's ecosystem similarly emphasizes human capital but through a more internationally oriented model. Mohamed bin Zayed University of Artificial Intelligence, founded in 2019, has recruited faculty internationally and built research connections extending into the United States (Rest of World, 2025, 2026). Canada has pursued a longer-running research-talent strategy through the Pan-Canadian AI Strategy and the CIFAR AI Chairs program, while more recent policy has emphasized attracting additional AI workers through immigration channels (CIFAR, 2026; Government of Canada, 2026). Singapore has the highest Talent score in the sample at 62.0, despite considerably lower R&D and Infrastructure scores. In its case, the data provide a clear picture of talent concentration, although the paper does not identify a single policy initiative sufficient to explain that score.

These cases reinforce the distinction between aggregate strength and breadth. India, Singapore, Canada, and the UAE are not simply smaller versions of the United States or China. Their ecosystems have developed around particular advantages, especially human capital, while other dimensions remain less developed. Such specialization can be a rational development strategy, particularly for countries that cannot match the capital or infrastructure expenditure of the largest economies. It also means, however, that aggregate rankings may conceal important differences in the structure and resilience of national AI capability.

8. Capability Is Not Sovereignty

The preceding analysis concerns the development of AI capability within a country. It does not establish that the country controls the technologies required to sustain that capability. This distinction has become increasingly important as governments use concepts such as “sovereign AI” to describe domestic investment in compute, data centers, models, and talent. Building national capacity can increase resilience and reduce some forms of dependence without eliminating dependence elsewhere in the technology stack.

The Center for New American Security's Sovereign AI Index illustrates the scale of this issue. Across more than 130 national AI infrastructure and model projects, approximately 70 percent involve at least one foreign technology partner, and roughly four-fifths of those projects involve a U.S. company. Nvidia accelerators appear in 52 percent of the projects tracked by the index (CNAS, 2026). The pattern extends across several layers of the AI stack, including advanced chips, server systems, cloud infrastructure, and networking.

This creates an important distinction between having AI capacity and controlling the technologies on which that capacity depends. A country can operate a national data center while relying on foreign accelerators; it can train domestic models while using foreign cloud infrastructure; and it can build a large domestic AI workforce while remaining dependent on foreign semiconductor and networking technologies. Domestic ownership of one layer therefore does not necessarily translate into autonomy across the system.

The distinction matters for interpreting the ecosystem scores used in this paper. A high Infrastructure score can indicate substantial domestic compute capacity without revealing who supplies the chips that power it. A high Talent score can indicate a strong concentration of AI-skilled workers without indicating control over the platforms on which those workers operate. Ecosystem strength and strategic autonomy should consequently be measured separately. The former concerns the development of national capability; the latter concerns dependence on external providers for the technologies required to exercise that capability.

9. Implications for AI Governance

The results suggest that national AI strategies should be evaluated using more than investment totals. The ten countries in this sample show that large differences in private investment can coexist with relatively similar levels of measured ecosystem strength. Spain, for example, reaches an ecosystem score of 27.9 with only \$542 million in AI venture investment, while Singapore attracts \$3.5 billion and scores 21.3. India combines substantial R&D and Talent capacity with investment far below that of the largest AI economies. These differences indicate that capital is better understood as one input into ecosystem development than as a direct measure of the capability that results from it.

The breadth analysis adds another layer to this conclusion. A national ecosystem can reach a given aggregate score through very different internal configurations, and those configurations matter for how the ecosystem can develop further. An ecosystem concentrated in talent may provide a strong foundation for future research and commercialization but remain constrained by weak infrastructure. An ecosystem concentrated in governance may have substantial institutional capacity without comparable research depth. A broad ecosystem, by contrast, contains stronger capabilities across several parts of the development chain. Reporting only an aggregate score therefore obscures the structure of national capability.

For policymakers, this suggests that national AI assessments should report both level and composition. The level of an ecosystem indicates how much capability has been developed; the composition indicates where that capability resides. Investment data can complement these measures by showing the scale of financial resources entering the ecosystem, but should not replace them. The distinction is particularly important when countries with very different development strategies are compared using a single headline number.

The sovereignty analysis adds a separate consideration. A country can possess high investment, strong research, significant talent, and substantial infrastructure while remaining dependent on foreign suppliers for critical components of the AI stack. If the policy objective is strategic autonomy rather than simply ecosystem growth, then measures of foreign technology dependence must be incorporated explicitly rather than inferred from domestic investment or capability scores.

The present analysis also points toward a natural extension of the research. A cross-sectional comparison cannot determine whether concentrated ecosystems subsequently broaden as investment increases, whether broad ecosystems are more successful at attracting investment, or whether countries deliberately specialize in particular dimensions. A longitudinal dataset could examine whether changes in investment precede changes in ecosystem breadth and whether different policy strategies produce distinct trajectories of capability development. The cross-sectional evidence presented here identifies the structural relationship that such an analysis should investigate.

10. Conclusion

AI investment has become one of the most common proxies for national AI strength, but the cross-country evidence examined here suggests that investment and capability should not be treated as interchangeable. Among the ten leading countries in Stanford HAI's 2024 Global AI Vibrancy ranking, AI investment has a positive but statistically insignificant association with the Non-Economic AI Ecosystem Score ($\rho = 0.41$, $p = 0.24$). The relationship is stronger when capability is measured by its distribution across dimensions: investment is negatively correlated with HHI ($\rho = -0.64$, $p = 0.048$), indicating that higher-investment countries in the sample tend to have more broadly distributed ecosystems.

The distinction between strength and breadth is important because aggregate scores can conceal substantially different models of development. The United States combines high investment with broad capability across the six dimensions, while China exhibits particularly strong R&D capability relative to its private investment. Spain's ecosystem is concentrated around policy and governance, while India, Singapore, the UAE, and Canada derive large portions of their scores from talent. These differences show

that countries can develop meaningful AI capability through different combinations of capital, institutions, research, talent, and policy.

The evidence does not establish that investment causes broader ecosystems. The relationship could reflect capital enabling diversification, stronger ecosystems attracting capital, or other factors influencing both. What the analysis does establish is that investment totals capture a different feature of national AI development from aggregate ecosystem strength, and that the relationship becomes more pronounced when capability is considered in terms of breadth.

Finally, neither investment nor ecosystem strength should be interpreted as evidence of strategic autonomy. Countries can build substantial domestic AI capabilities while remaining dependent on foreign firms for critical technologies. For governments and investors seeking to assess national AI strength, the relevant question is therefore not simply how much a country invests, but what that investment is building, how broadly capability is distributed across the ecosystem, and how dependent that capability remains on technologies controlled elsewhere.

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Data and Code

The Python code and dataset underlying the empirical analysis are available at:
<https://github.com/rachitamehta/AI-Investment-National-AI-Capability>